CBSE Test Paper 05

Chapter 6 Work Energy and Power

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1. Find the angle between force $F = 3\hat{i} + 4\hat{j} - 5\hat{k}$ and displacement $d = 5\hat{i} + 4\hat{j} + 3\hat{k}$. **1**
   a. $\theta = \cos^{-1}\left(\frac{8}{25}\right)$
   b. $\theta = \cos^{-1}\left(\frac{11}{25}\right)$
   c. $\theta = \sin^{-1}\left(\frac{11}{25}\right)$
   d. $\theta = \sin^{-1}\left(\frac{8}{25}\right)$
2. A 75.0-kg painter climbs a ladder that is 2.75 m long leaning against a vertical wall. The ladder makes an angle of 30° angle with the wall. How much work does gravity do on the painter? **1**
   a. -1950 J
   b. -1850 J
   c. -2050 J
   d. -1750 J
3. Balram pushes 500kg weight on a horizontal frictionless surface a distance of 10 m. The work done by gravitational force is **1**
   a. 200 J
   b. 5000 J
   c. 100 J
   d. 0 J
4. A 6.0-kg box moving at 3.0 m/s on a horizontal, frictionless surface runs into a light spring of force constant 75 N/cm Use the work–energy theorem to find the maximum compression of the spring. **1**
   a. 7.5 cm
   b. 8.5 cm
   c. 9.5 cm
   d. 6.5 cm
5. A 50.0-kg marathon runner runs up the stairs to the top of a 443-m-tall Tower. To lift herself to the top in 15.0 minutes, what must be her average power output? **1**
   a. 261 W

b. 221 W
c. 201 W
d. 241 W

6. When an air bubble rises in water, what happens to its potential energy? **1**
7. Two bodies of unequal mass are moving in the same direction with equal kinetic energy. The two bodies are brought to rest by applying retarding force of same magnitude. How would the distance moved by them before coming to rest compare? **1**
8. In which of the two types of collision i.e. elastic or inelastic, the momentum is conserved? What about K_E? **1**
9. An elevator which can carry a maximum load of 1800 kg (elevator + passenger) is moving up with a constant speed of 2 ms^{-1}. The frictional force opposing the motion is 4000 N. Determine the minimum power delivered by the motor to the elevator in watts as well as in horsepower. **2**
10. If momentum of a body increased by 300%, then what will be percentage increase in momentum of a body? **2**
11. A ball is dropped from the height h_1 and if it rebounds to a height h_2. Find the value of the coefficient of restitution? **2**
12. A long spring of spring constant 500 N/m is attached to a wall horizontally and surface below the spring is rough with coefficient of friction 0.75. A 100 kg mass block moving with a speed $10\sqrt{2}\text{ms}^{-1}$ strikes the spring. Find the maximum compression of the spring. (g =10 ms^{-2}). **3**
13. A ball of mass *m*, moving with a speed $2v_0$, collides inelastically (e > 0) with an identical ball at rest. Show that **3**
    i. For head-on collision, both the balls move forward.
    ii. For a general collision, the angle between the two velocities of scattered balls is less than 90°.
14. A car of mass 1000 kg accelerates uniformly from rest to a velocity of 54 km h^{-1} in 5 s. Calculate (i) its acceleration, (ii) its gain in kinetic energy, (iii) average power of the engine during this period. Neglect friction. **3**
15. Two identical steel cubes (masses 50 g, side 1cm) collide head-on face to face with a speed of 10 cm/s each. Find the maximum compression of each. Young's modulus for steel $Y = 2 \times 10^{11}$ N/m ^{2}. **5**